# week 1

① Sampling rate $f_s \geq 2f_m$, $f_s = \frac{1}{T_s}$

② quantisation:

$V_{pp} = 2V_p$, step size $q$, $-\frac{q}{2} \leq \text{error} \leq \frac{q}{2}$, $M(\text{levels}) = \frac{V_{PP}}{q}$, $M = 2^n$, $n$: bits

向上取2的整数倍

③ 误差功率 $P_Q = \frac{q^2}{12}$

④ SQNR:

① $SQNR = \frac{P_S}{P_Q} = \frac{\frac{M^2-1}{12}q^2}{\frac{q^2}{12}} = M^2 - 1 \approx M^2$

② $SQNR = 20\log_{10}M\,(dB) = 4.8 + 6n - \alpha_{dB}$, $\alpha_{dB} = \frac{V_p^2}{V^2}$ (mean value)

peak to mean ratio

⑤ bit 特率 $R_B = f_s \times N_B$ bps → minimum bandwidth $= \frac{R_B}{2}$

$N_B$: bits/symbol

∵ for binary, we can transmit up to 2 bits per Hz of bandwidth

⑥ Delta modulation

$\frac{\Delta}{T_s} \geq \max\left|\frac{dm(t)}{dt}\right|$

$T_s = \frac{1}{f_s}$

信号变化率

信号

week 2

① 概率: $P(A)(1-P(A|A)) = P(A|B)P(B) + P(A|C)P(C)$

$P(B)(1-P(B|B)) = P(B|A)P(A) + P(B|C)P(C)$

$P(C)(1-P(C|C)) = P(C|A)P(A) + P(C|B)P(B)$

$P(B:A) = P(A|B)P(B)$

② 信息和熵:

$I = \log_2(\frac{1}{p})$, $\log_2(x) = \frac{\log_{10}(x)}{\log_{10}(2)}$

$\begin{cases} \log_{10}2 = 0.3 \\ \log_{10}3 = 0.48 \\ \log_{10}5 = 0.70 \\ \log_{10}7 = 0.85 \end{cases}$

$H = \sum_i p_i \log_2(\frac{1}{p_i})$ bits/symbol

max entropy: 所有事件等概率 $\Rightarrow H_{max} = \log_2 N$ (N symbols)

information rate = $H \cdot R$ → prodnce seqnence rate

redundancy = $H_{max} - H$

③ source coding

大于symbol

$\log_2 x$ bits

Source efficiency: $\eta_{source} = \frac{H}{H_{max}}$

code efficiency: $\eta_{code} = \frac{H}{L}$ (average length) $\rightarrow \sum p_i n_i$ (bits number)

compression rate $r = \frac{\text{无编码长度}}{L}$ ?

④ PAM:

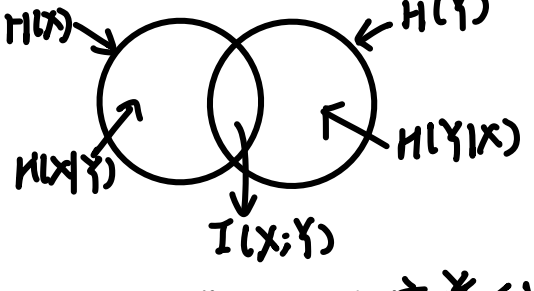
$R_b = \frac{1}{T_b}$ — bit interval

bit rate

$E_b = A^2 T_b$ 能量

白噪声 $n(t)$: 密度为 $\frac{1}{2}N_0$

⑤ 数字信道




$H(X,Y) = -\sum_{x,y} p(x,y) \log_2 p(x,y)$

$H(X) = -\sum_{x} p(x) \log_2 p(x)$

$H(X|Y) = -\sum_{x,y} p(x,y) \log_2 p(x|y)$

$H(Y|X) = H(X,Y) - H(X)$

$H(X|Y) = H(X,Y) - H(Y)$

$I(X;Y) = H(X) - H(X|Y) = H(Y) - H(Y|X) = \sum_{x,y} p(x,y) \cdot \log_2 \left( \frac{p(x,y)}{p(x)p(y)} \right)$

⑥ 二进制对称信道的信道容量

$$M = \frac{2^{NH(Y)}}{2^{NHb(\varepsilon)}} = 2^{N(H(Y)-Hb(\varepsilon))}$$

$$R = \frac{\log M}{N} = H(Y) - Hb(\varepsilon)$$ $M$ 个 input，每个长度为 $N$

$$C = R_{max} = 1 - Hb(\varepsilon)$$

## ⑦ AWGN信道

(channel capacity)

$$C_{bit/s} = W \log_2(1+SNR) = W\log\left(1+\frac{P}{N_0W}\right) = W\log\left(1+\eta\frac{E_B}{N_0}\right)$$

- $C_{bit/s}$：bit transmission rate
- $W$：bandwidth
- $P$：信号功率
- $SNR$：若dB，要换算位
- $\frac{N_0}{2}$ 噪声谱密度
- $P_N = N_0W$ 噪声功率
- $SNR = \frac{P}{P_N} = \frac{P}{N_0W}$

$$C_{bit/sym} = \frac{C_{bit/s}}{2W} = \frac{1}{2}\log(1+SNR) = \frac{1}{2}\log\left(1+\frac{P}{N_0W}\right)$$

$C_{bit/sym}$：symbol transmission rate

$\eta = \frac{R_B}{W}$ bit rate / bandwidth $\quad \eta_{max} = \frac{C_{bit/s}}{W} = \log(1+\frac{P}{N_0 W})$ bps/Hz

$P = E_B R_B \quad SNR = \frac{P}{P_N} = \frac{E_B R_B}{N_0 W} = \eta \frac{E_B}{N_0}$

P: 1s 耗能 $E_B$: 1bit 平均能耗

⑧ channel coding

原始长度 k，编码后长度 n，冗余 m = n-k

$R_C = \frac{k}{n}$ code rate $\quad$ transmission rate = $\frac{\text{produce rate}}{\text{code rate } R_C}$

线性分组码 code word + code word = code word

Hamming distance：$d_{min} \geq 2N_C+1$，$d_{min} \geq N_D+1$，码字中几个不同

Hamming weight：码字中几个1 $\quad$ $d_{min}$ = minimum weight

△ Generator Matrix G

$C = xG$

code word

$G = [I_k | P]$

parity matrix

identity matrix

parity check Matrix H

$H = [P^t | I_{n-k}]$ $k = p^t$的列数

⑨ Block decoding algorithm

$YH^t = cH^t + eH^t = eH^t$

syndrome: $S = eH^t$

code word $C = y + e$，e由$S = YH^t$确定

想要纠正错误，$2^m \geq n+1$

bits n+1 errors (包括全0情况)

x (information sequence) : first k bits of code word

△ code word 与 H 关系：

$H = \left[\begin{array}{cc|ccc} 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{array}\right]$ → $C = \begin{bmatrix} 00 & 0 & 0 & 0 \\ 01 & 0 & 1 & 1 \\ 10 & 1 & 0 & 0 \\ 11 & 0 & 1 & 1 \end{bmatrix}$ (P1 P2 P3)

take effect

(Hamming codes belong to the family of linear block codes. Hence, the minimum distance can be obtained as the minimum weight, where the weight of a code word is defined as the number of bits of value 1 in each sequence)